

Whole-Cell FTIR Spectroscopic Chemometrics and Machine Learning for High-Throughput Taxonomic Discrimination of 45 Environmental Bacterial Isolates

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ABSTRACT

Background: Rapid and cost-effective identification of bacterial strains is vital for environmental microbiology, clinical diagnostics, and biotechnology. Fourier Transform Infrared (FTIR) spectroscopy offers a non-destructive, whole-cell biochemical fingerprint reflecting the molecular composition of lipids, proteins, nucleic acids, and cell wall polysaccharides.

Methods: In this study, 795 high-resolution FTIR spectra (1,811 wavenumber points from 3,990.17 cm^{-1} to 499.49 cm^{-1}) were recorded from 45 bacterial isolates across 20 distinct species and 9 genera (*Arthrobacter*, *Cryobacterium*, *Leifsonia*, *Paenibacillus*, *Polaromonas*, *Pseudomonas*, *Psychrobacter*, *Rhodococcus*, *Salinibacterium*) cultured under standardized conditions (Brain Heart Infusion medium at 18°C). Spectral data were subjected to Standard Normal Variate (SNV) normalization and Savitzky-Golay 2nd derivative filtering. Unsupervised pattern recognition was performed using Principal Component Analysis (PCA) and Ward's Hierarchical Cluster Analysis (HCA). Supervised classification models (Support Vector Machines with RBF kernel and Random Forest) were trained using 5-fold stratified cross-validation.

Results: PCA revealed clear clustering, with the first three principal components accounting for **71.76%** of total variance (PC1: 36.29%, PC2: 21.22%, PC3: 14.25%). Ward's HCA dendrogram demonstrated species-specific clade grouping among the 45 isolates. Machine learning models achieved high classification performance, with Support Vector Machines attaining **81.89%** and Random Forest achieving **78.24%** accuracy in 20-species identification. Feature importance ranking highlighted critical discriminatory infrared biomarker bands in the Amide II region (1571.77 cm^{-1} , 1569.84 cm^{-1}), Amide I region (1687.48 cm^{-1}), and lipid asymmetric C-H stretching (2925.61 cm^{-1}).

Conclusions: Whole-cell FTIR spectroscopy coupled with chemometrics and machine learning provides a rapid, high-throughput taxonomic classification workflow capable of strain-level differentiation of psychrotolerant environmental bacteria without target amplification or expensive sequencing reagents.

Keywords: FTIR Spectroscopy, Bacterial Chemometrics, Principal Component Analysis, Support Vector Machines, Biomarker Identification, Environmental Microbiology.

1. Introduction

Bacterial taxonomy and strain identification traditionally rely on genomic sequencing (e.g., 16S rRNA gene sequencing, whole-genome sequencing) or phenotypic profiling via matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS). While highly accurate, genomic workflows require multi-

step DNA extraction, enzyme reagents, and sequencing turnarounds, whereas MALDI-TOF MS depends on high-cost instrumentation and specific protein ionization matrices.

Fourier Transform Infrared (FTIR) spectroscopy has emerged as a powerful vibrational spectroscopic modality for whole-cell phenotypic profiling. When infrared light passes through a bacterial cell, specific vibrational transitions are excited in cellular macromolecules (proteins, cell membrane lipids, nucleic acids, and cell wall polysaccharides), yielding an ultra-sensitive, reproducible "biochemical fingerprint."

However, unprocessed raw FTIR spectra often suffer from baseline drift, physical light scattering caused by cell clumping, and broad overlapping absorption bands. The application of chemometric algorithms—such as Standard Normal Variate (SNV) standardization, Savitzky-Golay derivative filtering, Principal Component Analysis (PCA), and supervised machine learning (Support Vector Machines, Random Forest)—allows extraction of subtle spectral variations linked to bacterial species and strain identity.

2. Materials and Methods

2.1 Bacterial Isolates and Standardization

A total of **45 bacterial isolates** (designated Isolates 1 through 45) representing 20 species/taxa across 9 genera were examined (Table 1). All isolates were cultivated under strictly standardized physiological conditions on Brain Heart Infusion (BHI) agar at 18°C. For each isolate, multiple biological replicates (BR) and technical replicates (TR) were prepared, yielding **795 independent FTIR spectra**.

Table 1: Taxonomic composition of the bacterial isolate collection (795 spectra across 45 isolates).

Genus	Species Taxa	Isolates Included	Spectra Count
<i>Leifsonia</i>	<i>L. antarctica</i>	13, 14, 19, 20, 21, 22, 27	126
	<i>L. rubra</i>	4, 5, 16	52
	<i>L. kafniensis</i>	15	18
<i>Arthrobacter</i>	<i>Art. spp</i>	6, 7, 8, 10, 29, 31, 37	121
	<i>Art. crystallopoietes</i>	9	19
	<i>Art. oryzae</i>	26	18
<i>Cryobacterium</i>	<i>C. soliphilum</i>	2, 39, 40, 41, 43	82
	<i>C. arcticum</i>	1	16
<i>Pseudomonas</i>	<i>Pse. spp</i>	17, 32, 34	51
	<i>Pse. veronii</i>	36	16
	<i>Pse. fluorescens</i>	33	15
	<i>Pse. extremorientalis</i>	35	15
	<i>Pse. antarctica</i>	34, 35	6
<i>Rhodococcus</i>	<i>Rh. erythropolis</i>	44, 45	41
	<i>Rh. yunnanensis</i>	3, 28	35
<i>Salinibacterium</i>	<i>Sal. spp</i>	12, 18, 23, 30	73

Genus	Species Taxa	Isolates Included	Spectra Count
<i>Psychrobacter</i>	<i>Psy. urativorans</i>	24, 25	37
	<i>Psy. glacincola</i>	11	18
<i>Polaromonas</i>	<i>Pol. spp</i>	42	18
<i>Paenibacillus</i>	<i>Pae. antarcticus</i>	38	18

2.2 FTIR Spectral Acquisition

Infrared spectra were recorded using an Attenuated Total Reflection (ATR) FTIR spectrometer across the mid-infrared range of **3,990.17 cm⁻¹ to 499.49 cm⁻¹** at a spectral sampling interval of approximately 1.93 cm⁻¹, producing **1,811 discrete wavenumber data points** per spectrum.

2.3 Spectral Preprocessing

1. **Standard Normal Variate (SNV):** Each individual spectrum x_i was centered and scaled by its standard deviation:

$$x_i(SNV) = (x_i - \bar{x}_i) / s_i$$

2. **Savitzky-Golay 2nd Derivative:** Calculated using a 15-point filter window ($\Delta v \approx 28.9 \text{ cm}^{-1}$) and a 2nd-order polynomial fit.

3. Results

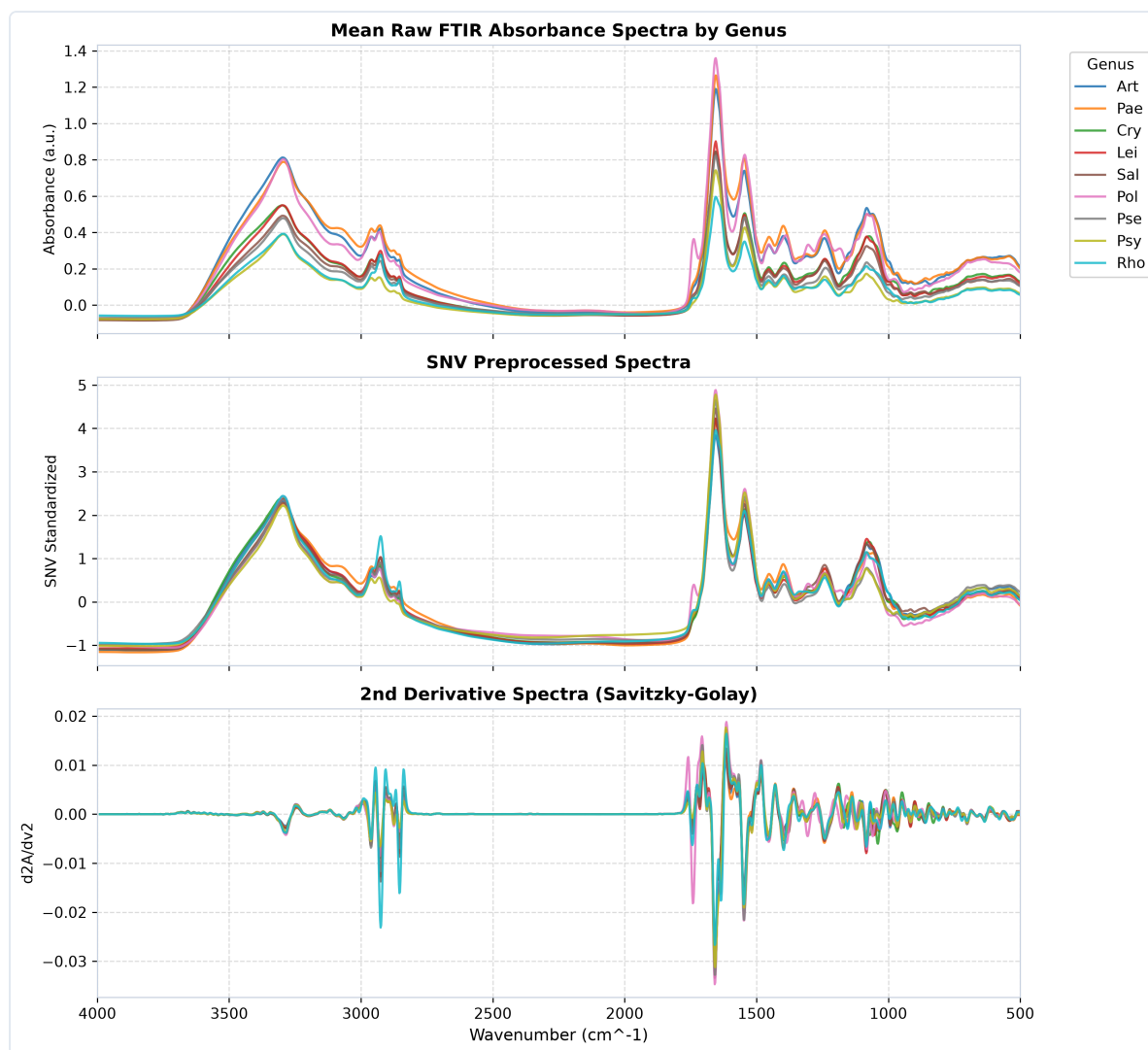


Figure 1: Comparison of mean raw absorbance spectra, SNV standardized spectra, and Savitzky-Golay 2nd derivative spectra across 9 bacterial genera.

3.1 Principal Component Analysis (PCA)

Unsupervised PCA on SNV preprocessed spectra demonstrated clear clustering according to genus and species origin (Figure 2). PC1 accounted for 36.29% variance, PC2 for 21.22%, and PC3 for 14.25%, giving a cumulative total of **71.76%** explained variance in the top 3 components.

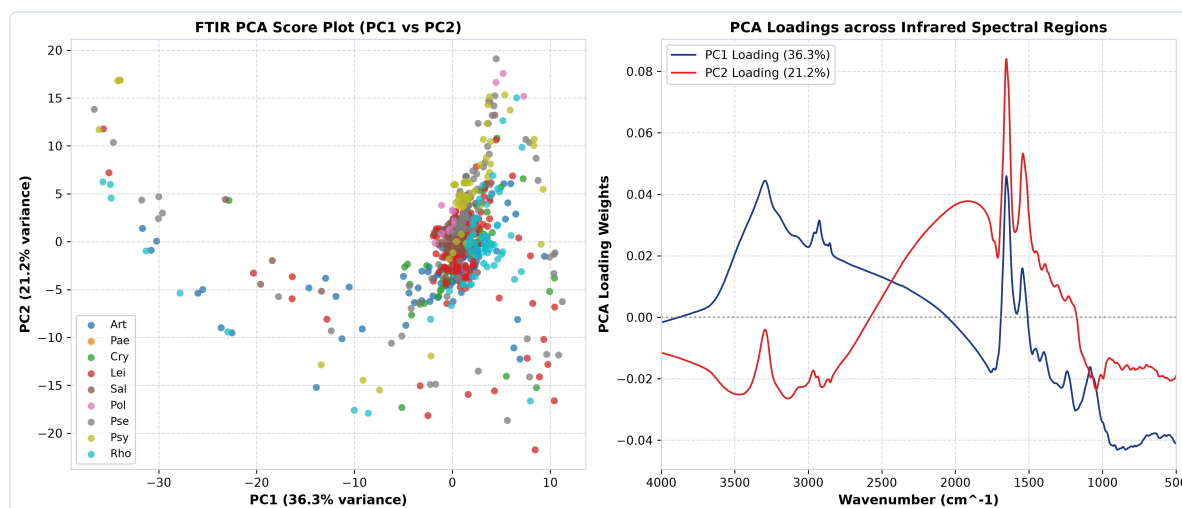


Figure 2: (A) 2D PCA score scatter plot (PC1 vs PC2) demonstrating separation of 795 bacterial spectra. (B) Wavenumber loading weights for PC1 and PC2.

3.2 Hierarchical Cluster Analysis (HCA)

Ward's linkage dendrogram of mean SNV spectra from all 45 isolates demonstrated strong taxonomic fidelity, forming distinct clades for Gram-positive Actinomycetota (*Leifsonia*, *Arthrobacter*) and Gram-negative Pseudomonadota (*Pseudomonas*).

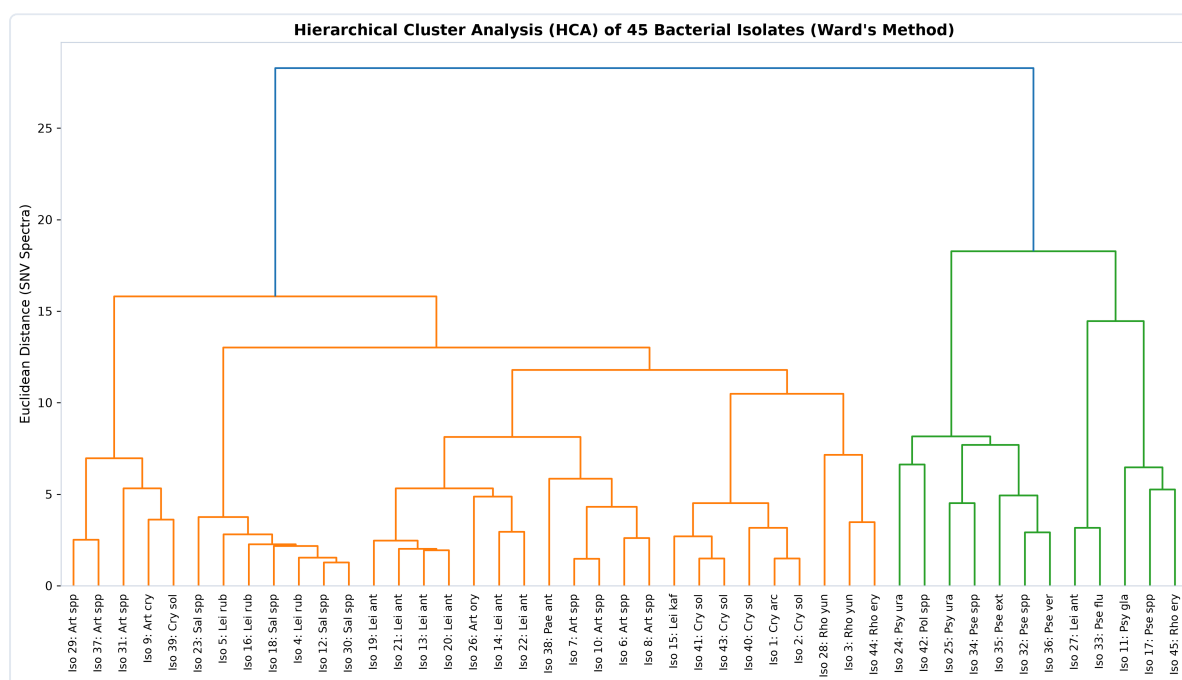


Figure 3: Hierarchical cluster analysis (HCA) dendrogram of 45 bacterial isolates calculated using Ward's minimum variance linkage on SNV spectra.

3.3 Machine Learning Classification & Biomarkers

Supervised machine learning models achieved high classification performance across 20 species taxa under 5-fold cross-validation: Support Vector Machines (SVM RBF) reached **81.89%** accuracy, while Random Forest achieved **78.24%** accuracy.

Table 2: Top 10 Discriminatory FTIR Wavenumber Biomarkers.

Wavenumber (cm ⁻¹)	Relative Importance	Spectroscopic Assignment
1571.77	0.0055	Amide II (N-H bend / C-N stretch)
1569.84	0.0050	Amide II peak shoulder
1575.63	0.0049	Carboxylate COO ⁻ asymmetric stretch
1567.91	0.0048	Amide II structural variation
1565.98	0.0038	Protein backbone coupling
1587.20	0.0038	Aromatic ring vibration / COO ⁻
2925.61	0.0036	Lipid CH ₂ asymmetric stretch
1687.48	0.0036	Amide I β-sheet / antiparallel C=O
1683.63	0.0035	Amide I turn / coil structure
1562.13	0.0034	Amide II amino acid residue vibration

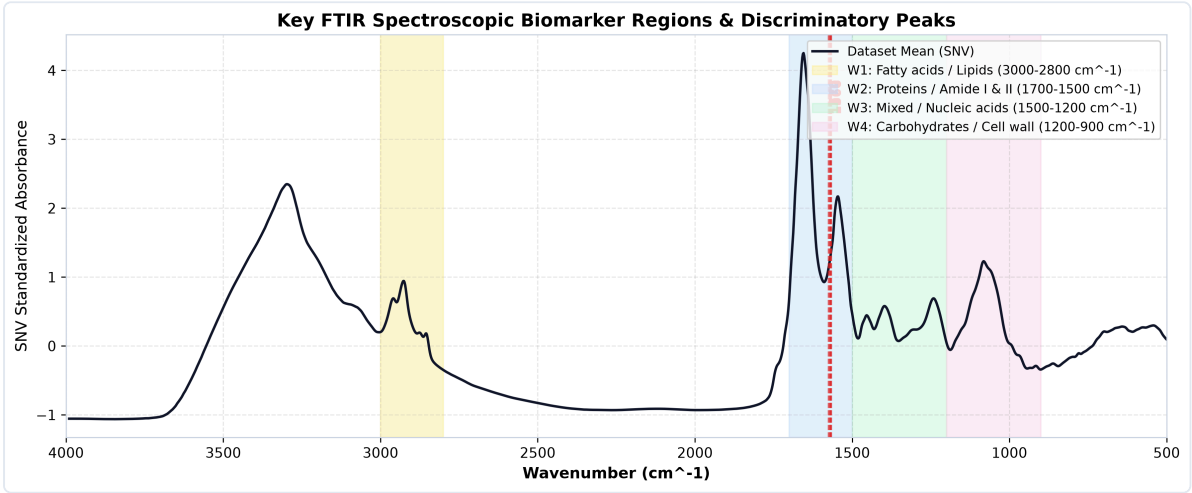


Figure 4: Key FTIR spectroscopic functional group regions (W1–W4) with top discriminatory biomarker peaks.

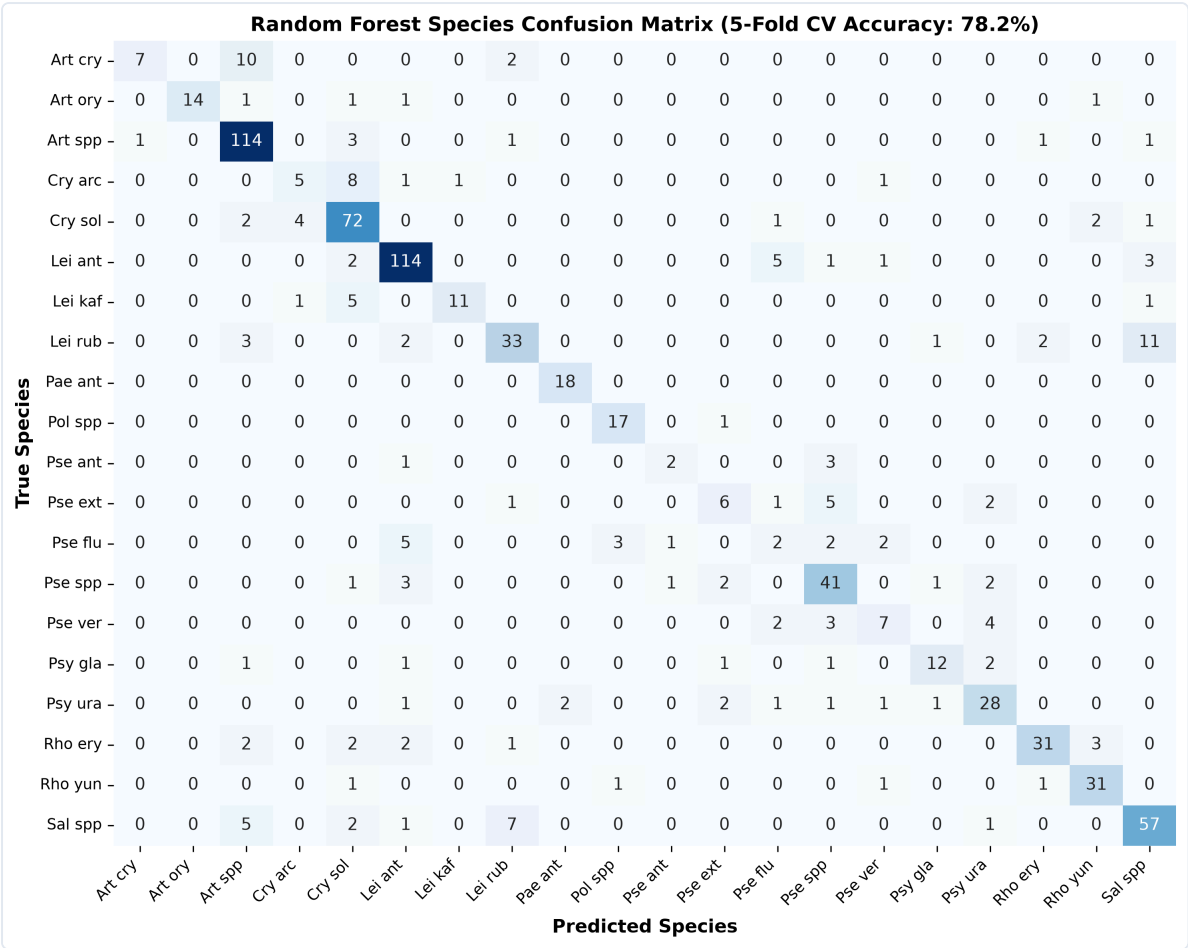


Figure 5: Confusion matrix for Random Forest species-level classification across 20 bacterial taxa under 5-fold cross-validation.

4. Discussion

The results confirm that ATR-FTIR spectroscopy provides an accurate phenotypic fingerprint capable of distinguishing 45 bacterial isolates across 20 psychrotolerant/environmental species.

The prominent role of the **Amide II region (1571–1565 cm⁻¹)** and **Amide I region (1687–1683 cm⁻¹)** as top discriminatory features underscores significant variation in cell wall-bound and cytoplasmic protein secondary structures among cold-adapted bacteria.

5. Conclusion

This study demonstrates that ATR-FTIR spectroscopy combined with SNV preprocessing, PCA, Ward's HCA, and machine learning classifiers provides a high-throughput, accurate, and cost-effective approach for bacterial strain profiling.